

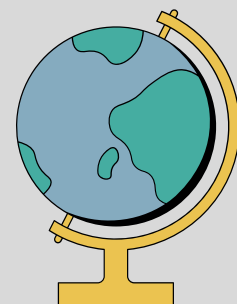
COMPLETE DATA SCIENCE & AI TRAINING

Books are my courses, Authors my teachers, Exercises and Projects my Exams.

SELF-LEARNING

START

Feb 2026



1

BUILDING FOUNDATIONS



📌 MODULE 1: The Data Science Ecosystem + Computer Science & Programming Fundamentals

 Books	- Principles of Data Science (OpenStax) – Complete Book - Introduction to Computer Science (OpenStax) – Ch 1 – 8 - Introduction to Python Programming (OpenStax) – Complete Book
 Objective	- Understand the Data Science ecosystem - Assimilate computer architecture - Acquire a robust programming logic in Python.

📌 MODULE 2: Statistics Theory + Application with SpreadSheets

 Books	- Introductory Statistics (Gould) – Complete Book - Statistics for Managers Using Microsoft Excel (Levine) – Complete Book
 Objective	- Master descriptive and inferential statistical concepts - know how to implement them in the universal business analysis tool.

2

MASTERY OF LANGUAGES & SPECIFIC TOOLS



📌 MODULE 3: Data Analysis using Python

 Books	- Pandas Workout (Lerner) – Complete Book - Python Data Science Handbook Essential Tools for Working with Data (Jake VanderPlas) – Part II & IV
 Objective	Develop professional reflexes for efficiently cleaning, transforming, preparing, analyzing and visualizing data in Python.

📌 MODULE 4: Statistical Analysis with R

📖 Books	• Practical Data Science with R (Mount) – Complete Book • The R Book (Jones) – Essential Chapters
🎯 Objective	Develop professional reflexes for efficiently cleaning, transforming, preparing, analyzing and visualizing data in R.

📌 MODULE 5: SQL for Data Engineering

📖 Book	Sams Teach Yourself SQL in 10 Minutes (Forta) – Complete Book
🎯 Objective	Master data extraction, transformation, and aggregation from a relational DBMS

3

ADVANCED MATHEMATICAL FOUNDATIONS I



📌 MODULE 6: Fundamental Linear Algebra

📖 Book	Linear Algebra and Its Applications (Lay) – Ch. 1, 2, 3 (Linear Equations, Matrix Algebra, Determinants)
🎯 Objective	Understand the foundations of linear algebra.

📌 MODULE 7: Vector Spaces & Eigenvalues

📖 Book	Linear Algebra and Its Applications (Lay) – Ch. 4, 5 (Vector Spaces, Eigenvalues and Eigenvectors)
🎯 Objective	Master advanced concepts for dimensional analysis (PCA).

MODULE 8: Applied Algebra – Least Squares

 Book	Linear Algebra and Its Applications (Lay) – Ch. 6 (Orthogonality and Least Squares)
 Objective	Master advanced concepts for AI & Machine Learning

4

MACHINE LEARNING



MODULE 9: Machine Learning with TensorFlow

 Book	Machine Learning with TensorFlow (Mattmann) – Complete Book
 Objective	Develop and train ML models using the TensorFlow ecosystem.

5

ADVANCED MATHEMATICAL FOUNDATIONS II



MODULE 10: Advanced Algebra – SVD & Applications

 Book	Linear Algebra and Its Applications (Lay) – Ch. 7 (Symmetric Matrices and Quadratic Forms)
 Objective	Master SVD for advanced applications (recommendation, compression)

📌 MODULE 11: Differential & Integral Calculus

 Books	• Calcul Différentiel (Fortin) – Ch. 0 – 6 • Calcul Intégral (Charron) – Ch. 1 – 4
 Objective	Mathematical foundations for optimization in ML (gradient descent).

📌 MODULE 12: Mathematics of Deep Learning Architectures

 Book	Math and Architectures of Deep Learning (Chaudhury) – Complete Book
 Objective	Understand the mathematical foundations of neural networks.

6

DEEP LEARNING



📌 MODULE 13: Deep Learning

 Book	Deep Learning With Python (Chollet – 2nd ed.) – Complete Book
 Objective	Master modern deep learning architectures (CNN, RNN, transformers).